X-Ray Powder Diffraction

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1 Abstract

X-ray diffraction (XRD) is a powerful technique widely employed in materials science to investigate the structural properties of crystalline materials. One crucial aspect of this analysis is the determination of crystallite size, which plays a pivotal role in understanding material properties and behavior. This report explores the theory and methodologies behind estimating the average crystallite size using the peak broadening observed in XRD data, and interpreting the results while highlighting factors influencing the accuracy.

2 Resources used

- Powder XRD data of a sample with known crystallite size and an unknown sample whose crystallite size has to be determined.
- Origin software

3 Theory

Bragg's law helps to determine the angles for coherent scattering of waves from a large crystal lattice by describing the relationship between the angle between the incident X-ray beam and the crystallographic plane (θ), the wavelength of X-rays (λ), and the spacing between the crystallographic planes (d_{hkl}) in a crystalline material:

$$n\lambda = 2d_{hkl}sin\theta \tag{1}$$

where n is an integer (the order of the diffraction peak).

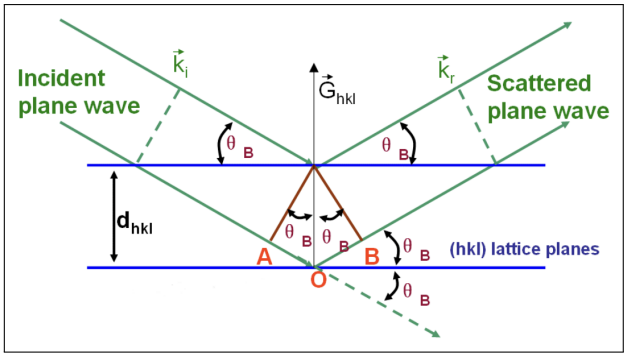


Figure 1: Illustration of Bragg's law

One of the primary applications of XRD is the determination of crystallite size, which refers to the dimensions of individual crystalline domains within a material. Crystallite size can significantly affect material properties such as mechanical strength, electrical conductivity, and thermal stability, making its accurate estimation crucial in various fields, including materials science, nanotechnology, and metallurgy.

Powder XRD is a widely used technique for analyzing powdered or polycrystalline materials. Unlike single-crystal XRD, which requires a well-ordered, single-crystal sample, powder XRD can analyze a wide range of materials, including minerals, metals, etc. In XRD analysis, the broadening of diffraction peaks is primarily caused by two factors: instrumental broadening and crystallite size broadening. Instrumental broadening is due to the limitations of the X-ray source, optics, and detectors, and it can be minimized through equipment calibration and optimization. On the other hand, crystallite size broadening arises from the finite size of crystalline domains in the material.

Factors contributing to the peak broadening are:

- Instrumental broadening
- Crystallite size
- Microstrain
- Solid solution inhomogeneity
- Temperature factors

The peak profile is a convolution of the profiles from all of these contributions. We will be using the contribution for broadening from Instrument, crystallite size, and microstrain. The Scherrer equation is a fundamental tool for estimating crystallite size based on XRD data. It relates the broadening of diffraction peaks to the average crystallite size using the following equation:

$$D = \frac{K\lambda}{\beta cos\theta} \tag{2}$$

where,

- D is the average crystallite size,
- K is the Scherrer constant, which depends on the shape of the crystallite (typically assumed to be around 0.94 rad),
- λ is the wavelength of the X-ray source,
- β is the full width at half-maximum (FWHM) of the diffraction peak in radians,
- θ is the Bragg angle.

If β_{total} is the total Full-Width-Half-Maximum peak broadening of an XRD peak, then we have:

$$\beta_{total} = \beta_{crystal} + \beta_{inst} + \beta_{strain} \tag{3}$$

where
- $\beta_{crystal}$ is the peak broadening due to the crystallite sample.
- β_{inst} is instrumental peak broadening.
- β_{strain} is peak broadening due to microstrain in the crystal.

Scherrer's formula for $\beta_{crystal}$ for a peak at 2θ is given by:

$$\beta_{crystal} = \frac{K\lambda}{Dcos(\theta)} \tag{4}$$

For microstrain broadening, we have:

$$\beta_{strain} = 4\epsilon\, tan\theta \tag{5}$$

where ϵ is the **strain parameter**.

4 Methodology

By estimating the broadening from a single peak using a known sample and either a Gaussian or a Lorentzian function which fits the best with the data neglecting the contribution from the lattice strain, β_{inst} can be found using: $\beta_{inst} = \beta_{total}^{known} - \beta_{crystal}^{known}$.

Next, 5-6 peaks in the XRD data of the unknown sample are analyzed and then d is determined using Scherrer's formula without taking microstrain into account. Subsequently, strain is taken into account by plotting a Williamson-Hall plot and then the strain parameter ϵ is estimated from the plot.

4.1 Analysis of known sample (Measurement of $\beta_{\text{instrument}}$)

For this case we will ignore the β_{strain} part and just use the rest of two β. As β_{total} can be obtained from FWHM of the XRD peaks and $\beta_{crystal}$ can be obtained from (4) for known a D.

Given Data sets:

- $K = 0.94$
- $\lambda = 1.5403$Å
- $D = 40nm$

4.1.1 Gaussian Fit method

For using Gaussian fit, we have the relation,

$$\beta_{total}^2 = \beta_{inst}^2 + \beta_{crystal}^2 \tag{6}$$

4.1.2 Lorentz fit

For using Gaussian fit, we have the relation,

$$\beta_{total} = \beta_{crystal} + \beta_{inst} \tag{7}$$

4.1.3 Plot

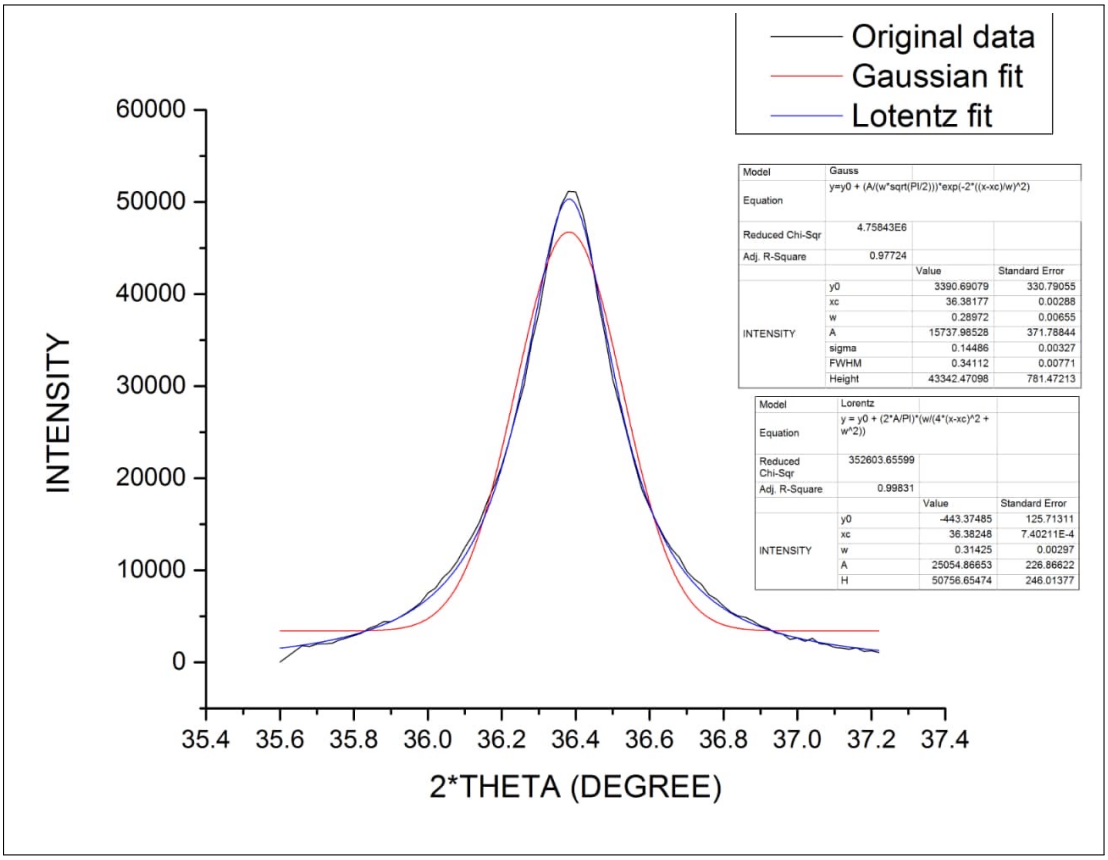


Figure 2: Fitting a prominent peak of the XRD data for the known sample

4.1.4 Data

Type	θ_{rad}	β_{cryst}	$(fwhm)°$	β_{total}	β_{inst}
Gaussian	0.317490837	0.003810871	0.34112	0.005953667	0.004574212
lorentz	0.317497033	0.003810878	0.31425	0.005484697	0.001673819

4.1.5 Conclusion

From the above plot we can see the Lorentz fit has adapted the curve most accurately.
So,

$$\boxed{\boldsymbol{\beta_{instrument} \approx 0.001673819 \text{ rad}}} \tag{8}$$

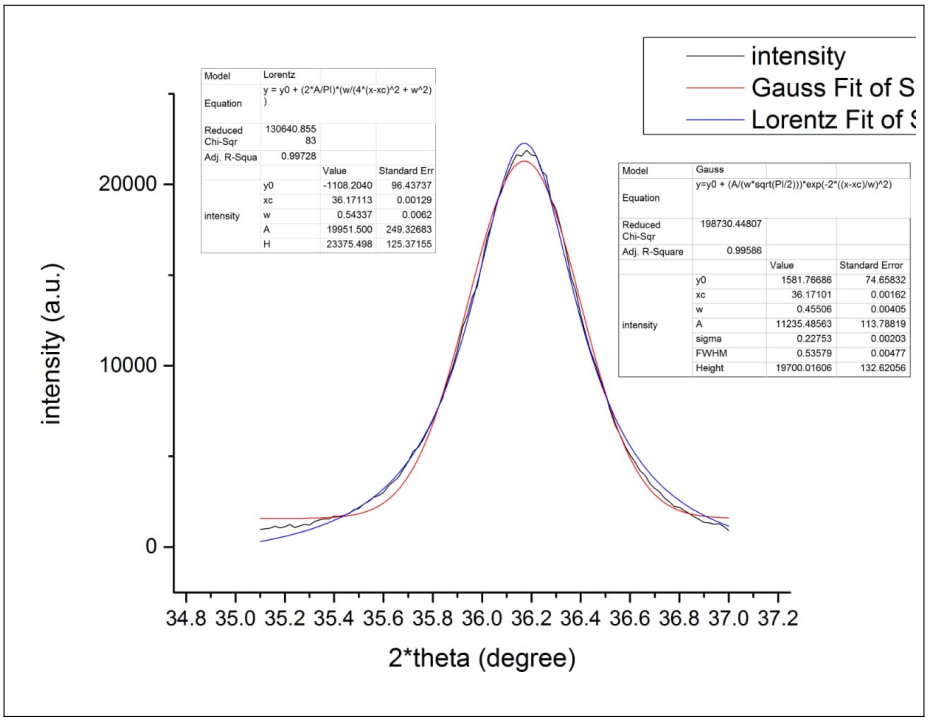
4.2 Analysis for Unknown Sample (Crystallite size Estimation)

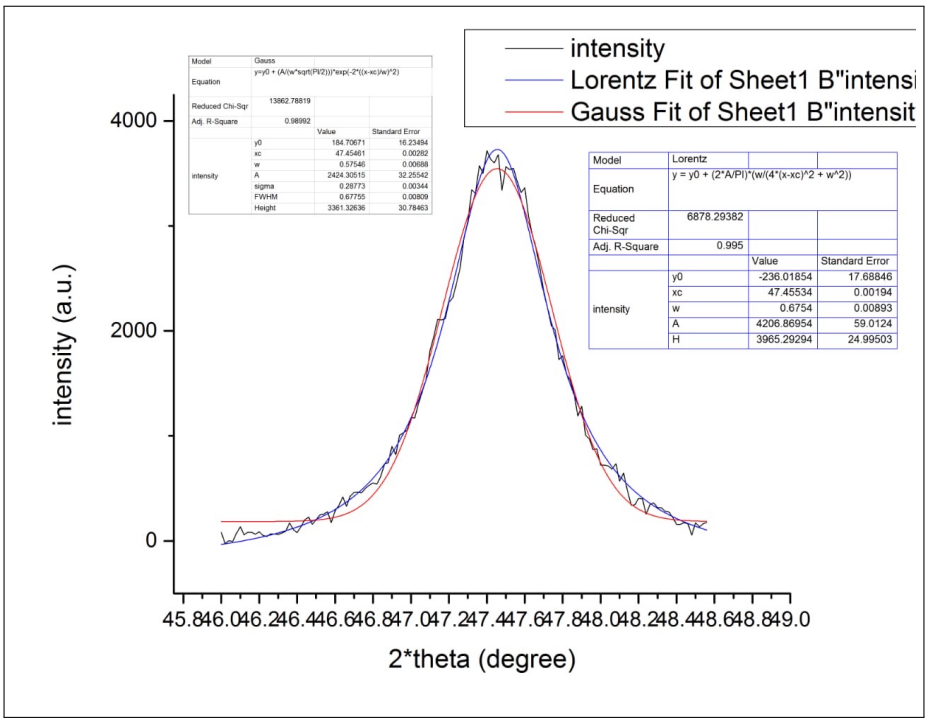
For this scenario, we will ignore the β_{strain} part.
Steps:

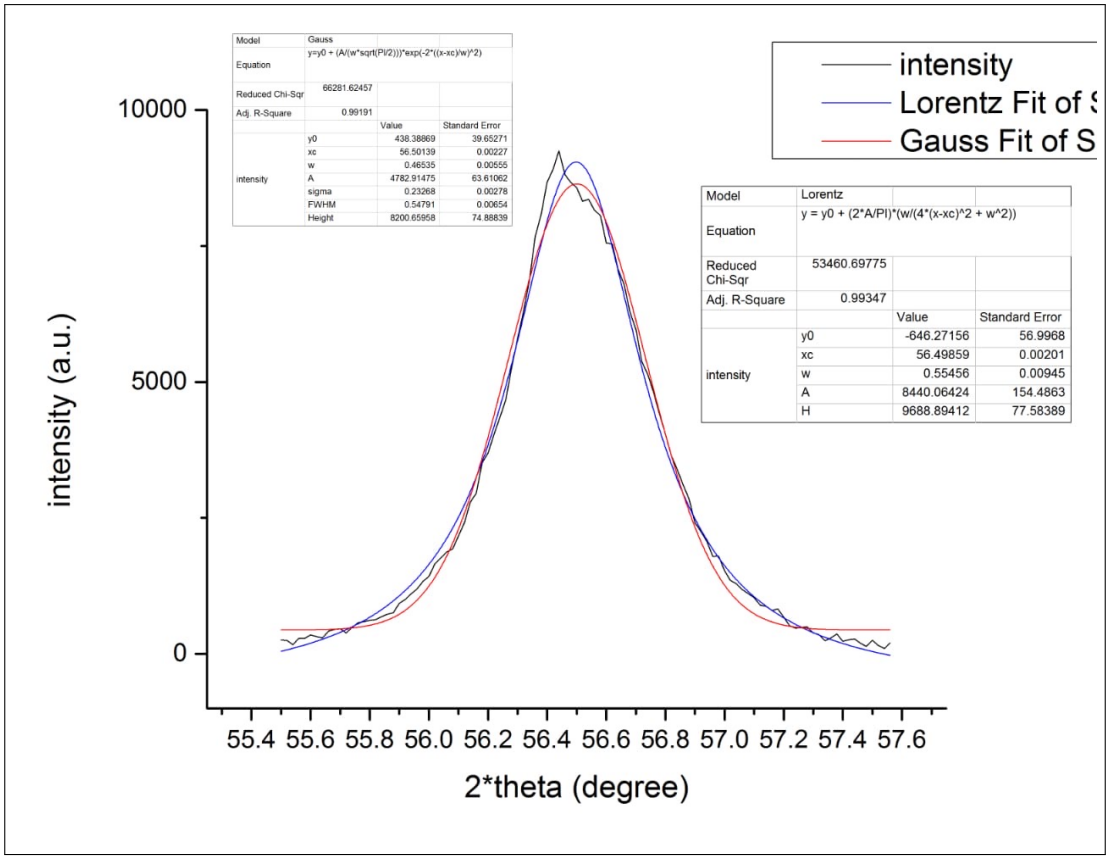
- Analyze different different peaks (7-8 peaks) of given sample
- Obtain FWHM to get β_{total} and θ(at the angle XRD Maxima occurs)
- Use β_{inst} and β_{total} on equations (6) & (7) to get β_{cryst}

- to obtain D from (4) using β_{cryst} and θ
- Repeat same for the other peaks
- Discard unusual data
- Average out to get D_{avg}

Plot

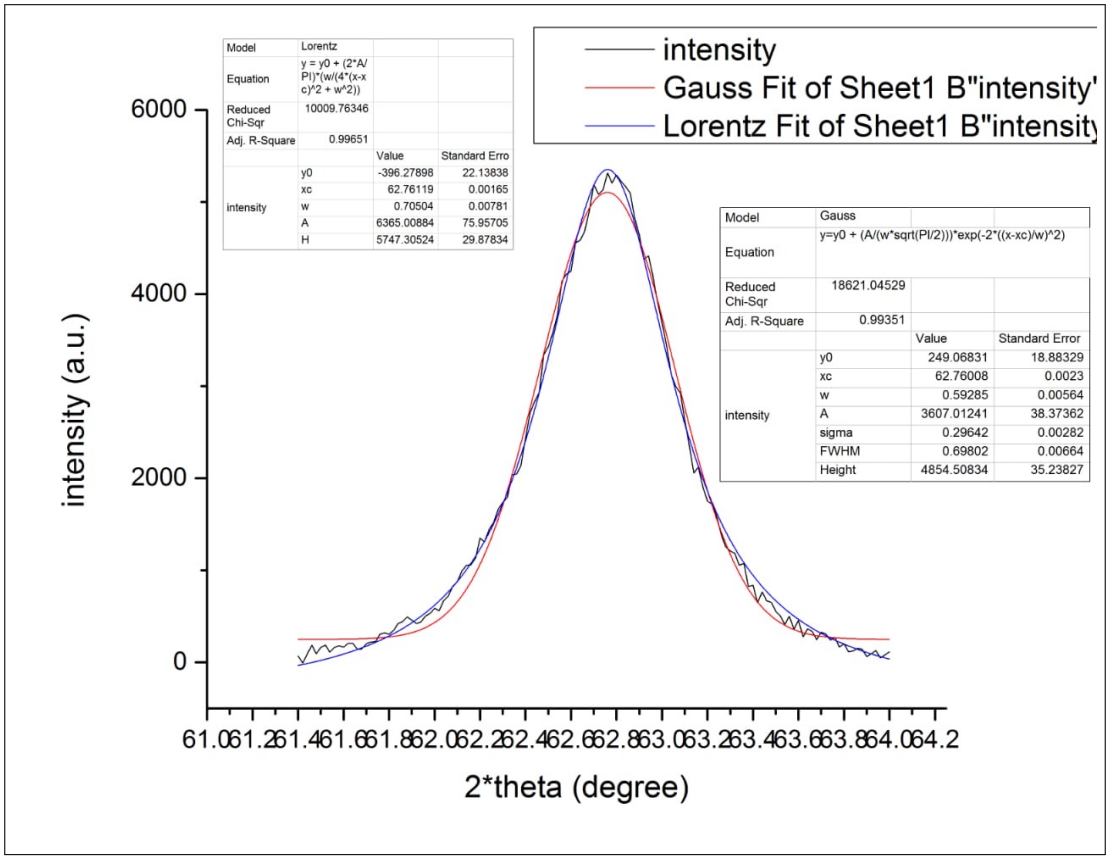






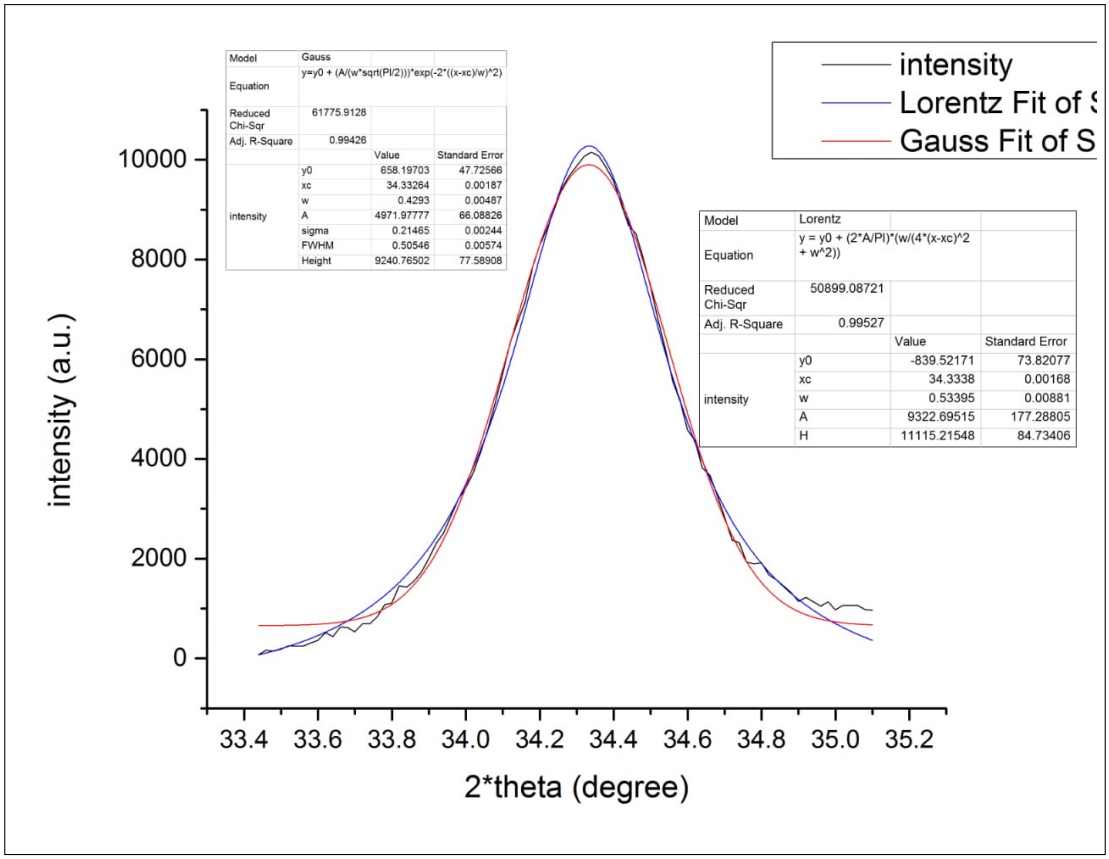
Model	Gauss		
Equation	y=y0 + (A/(w*sqrt(PI/2)))*exp(-2*((x-xc)/w)^2)		
Reduced Chi-Sqr	66281.62457		
Adj. R-Square	0.99191		
		Value	Standard Error
intensity	y0	438.38869	39.65271
	xc	56.50139	0.00227
	w	0.46535	0.00655
	A	4782.91475	63.61062
	sigma	0.23268	0.00278
	FWHM	0.54791	0.00654
	Height	8200.65958	74.88639

Model	Lorentz		
Equation	y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))		
Reduced Chi-Sqr	53460.69775		
Adj. R-Square	0.99347		
		Value	Standard Error
intensity	y0	-646.27156	56.9968
	xc	56.49859	0.00201
	w	0.55456	0.00945
	A	8440.06424	154.4863
	H	9688.89412	77.58389



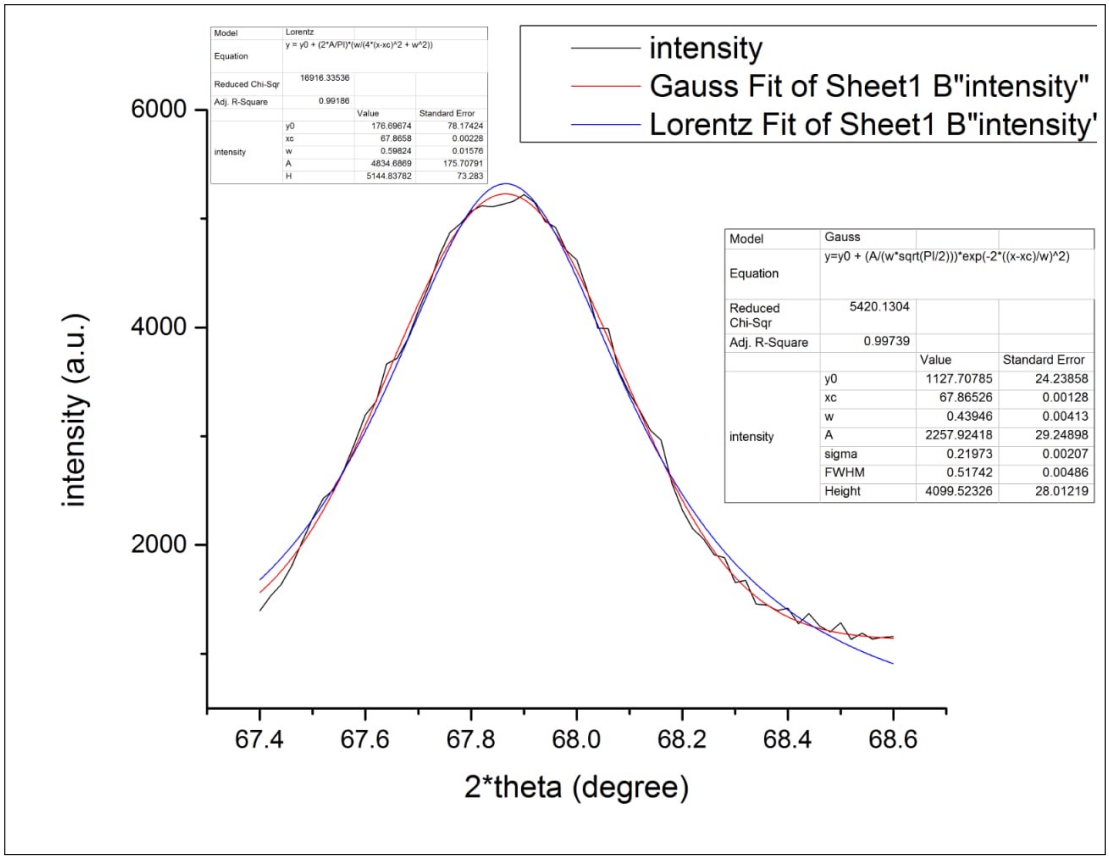
Model	Lorentz		
Equation	y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))		
Reduced Chi-Sqr	10009.76346		
Adj. R-Square	0.99651		
		Value	Standard Erro
intensity	y0	-396.27898	22.13838
	xc	62.76119	0.00165
	w	0.70504	0.00781
	A	6365.00884	75.95705
	H	5747.30524	29.87834

Model	Gauss		
Equation	y=y0 + (A/(w*sqrt(PI/2)))*exp(-2*((x-xc)/w)^2)		
Reduced Chi-Sqr	18621.04529		
Adj. R-Square	0.99351		
		Value	Standard Error
intensity	y0	249.06831	18.88329
	xc	62.76008	0.0023
	w	0.59285	0.00564
	A	3607.01241	38.37362
	sigma	0.29642	0.00282
	FWHM	0.69802	0.00664
	Height	4854.50834	35.23827



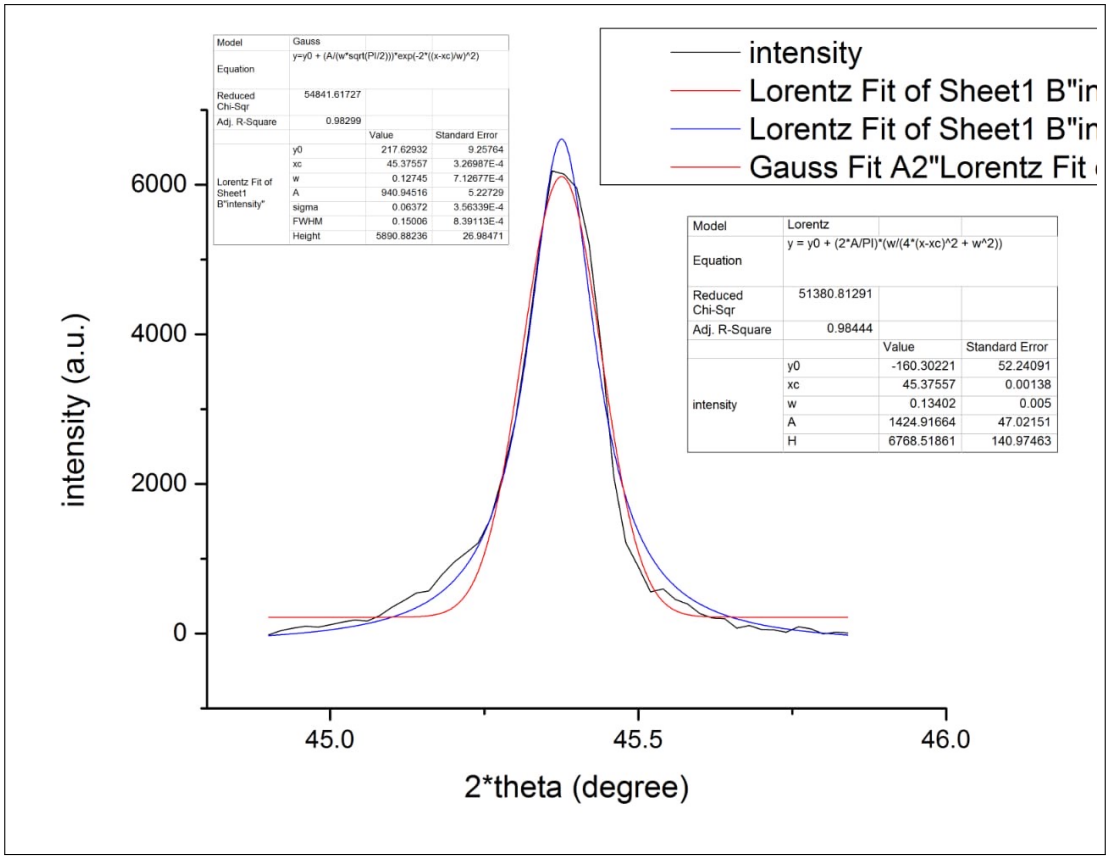
Model	Gauss		
Equation	y=y0 + (A/(w*sqrt(PI/2)))*exp(-2*((x-xc)/w)^2)		
Reduced Chi-Sqr	61775.9128		
Adj. R-Square	0.99426		
		Value	Standard Error
intensity	y0	658.19703	47.72566
	xc	34.33264	0.00187
	w	0.4293	0.00487
	A	4971.97777	66.08826
	sigma	0.21465	0.00244
	FWHM	0.50546	0.00574
	Height	9240.76502	77.58908

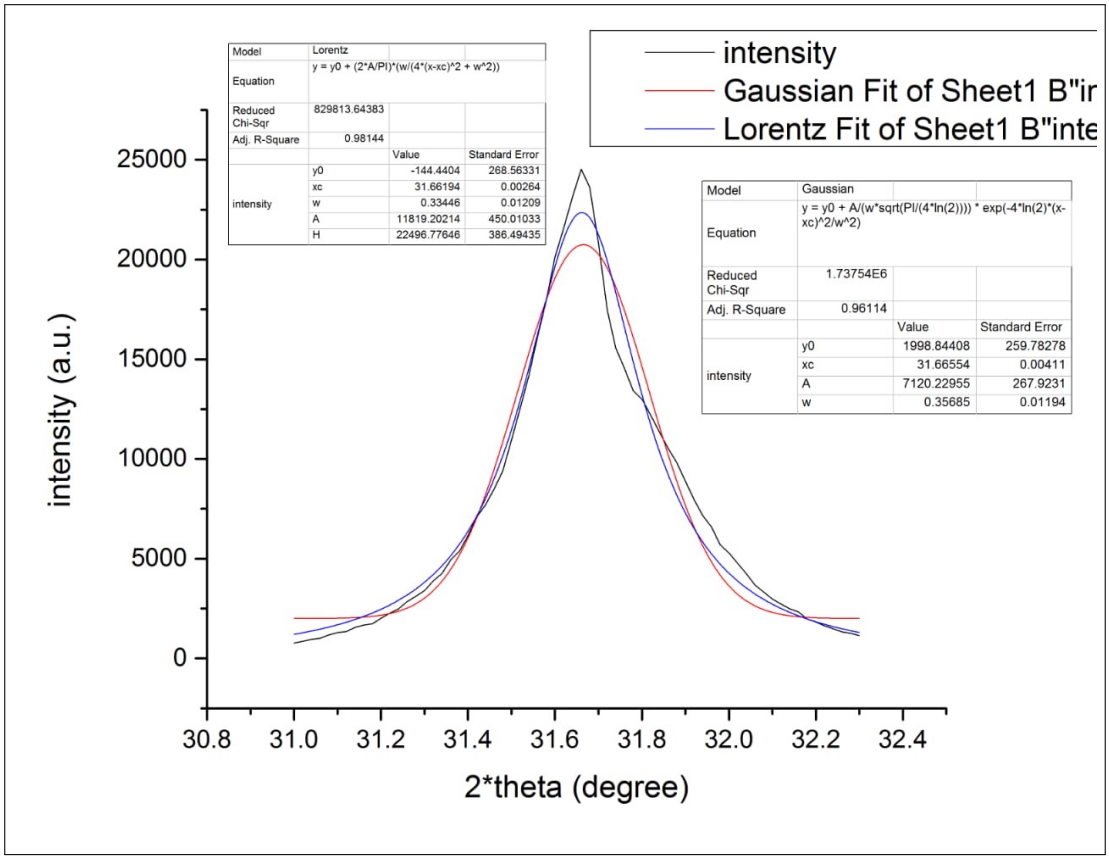
Model	Lorentz		
Equation	y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))		
Reduced Chi-Sqr	50899.08721		
Adj. R-Square	0.99527		
		Value	Standard Error
intensity	y0	-839.52171	73.82077
	xc	34.3338	0.00168
	w	0.53395	0.00881
	A	9322.69515	177.28805
	H	11115.21548	84.73406



Model	Lorentz		
Equation	y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))		
Reduced Chi-Sqr	16916.33536		
Adj. R-Square	0.99186		
		Value	Standard Error
intensity	y0	176.69674	78.17424
	xc	67.8658	0.00228
	w	0.59824	0.01576
	A	4834.6869	175.70791
	H	5144.83782	73.283

Model	Gauss		
Equation	y=y0 + (A/(w*sqrt(PI/2)))*exp(-2*((x-xc)/w)^2)		
Reduced Chi-Sqr	5420.1304		
Adj. R-Square	0.99739		
		Value	Standard Error
intensity	y0	1127.70785	24.23858
	xc	67.86526	0.00128
	w	0.43946	0.00413
	A	2257.92418	29.24898
	sigma	0.21973	0.00207
	FWHM	0.51742	0.00486
	Height	4099.52326	28.01219


intensity
Lorentz Fit of Sheet1 B"ir
Lorentz Fit of Sheet1 B"ir
Gauss Fit A2"Lorentz Fit
Model Gauss
Equation y=y0 + (A/(w*sqrt(PI/2)))*exp(-2*((x-xc)/w)^2)
Reduced Chi-Sqr 54841.61727
Adj. R-Square 0.98299
Value Standard Error
Lorentz Fit of Sheet1 B"intensity"
y0 217.62932 9.25764
xc 45.37557 3.26987E-4
w 0.12745 7.12677E-4
A 940.94516 5.22729
sigma 0.06372 3.56339E-4
FWHM 0.15006 8.39113E-4
Height 5890.88236 26.98471
Model Lorentz
Equation y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))
Reduced Chi-Sqr 51380.81291
Adj. R-Square 0.98444
Value Standard Error
intensity
y0 -160.30221 52.24091
xc 45.37557 0.00138
w 0.13402 0.005
A 1424.91664 47.02151
H 6768.51861 140.97463
intensity (a.u.)
6000
4000
2000
0
45.0
45.5
46.0
2*theta (degree)


intensity
Gaussian Fit of Sheet1 B"ir
Lorentz Fit of Sheet1 B"inte
Model Lorentz
Equation y = y0 + (2*A/PI)*(w/(4*(x-xc)^2 + w^2))
Reduced Chi-Sqr 829813.64383
Adj. R-Square 0.98144
Value Standard Error
intensity
y0 -144.4404 268.56331
xc 31.66194 0.00264
w 0.33446 0.01209
A 11819.20214 450.01033
H 22496.77646 386.49435
Model Gaussian
Equation y = y0 + A/(w*sqrt(PI/(4*ln(2)))) * exp(-4*ln(2)*(x-xc)^2/w^2)
Reduced Chi-Sqr 1.73754E6
Adj. R-Square 0.96114
Value Standard Error
intensity
y0 1998.84408 259.78278
xc 31.66554 0.00411
A 7120.22955 267.9231
w 0.35685 0.01194
intensity (a.u.)
25000
20000
15000
10000
5000
0
30.8
31.0
31.2
31.4
31.6
31.8
32.0
32.2
32.4
2*theta (degree)

Data

4.2.1 Gaussian data

Equation (6) used for analysis. Also we used $\boldsymbol{\beta_{instrument}^{gaussian} \approx 0.00457\ rad}$ for the analysis.

$(fwhm)^\circ$	β_{total}	β_{cryst}	θ	$D(nm)$
0.35685	0.006228207	0.004226955	0.276333966	35.61123064
0.53579	0.0093513	0.008156187	0.315651609	18.67821475
0.67755	0.011825478	0.010904977	0.414119595	14.50602547
0.51742	0.009030683	0.007786515	0.592236117	22.41591228
0.50546	0.008821941	0.007543423	0.299608804	20.09279725
0.69802	0.012182747	0.011291409	0.547685017	15.02269769
0.54791	0.009562834	0.008397879	0.493067644	19.57622431
0.15006	0.002619041	-	0.395976548	-

here we can see clearly at $\theta \approx 0.3960$ we are getting $\beta_{total} \approx 0.00262\ rad$ whereas $\boldsymbol{\beta_{instrument}^{gaussian} \approx 0.00457\ rad}$. So, for further calculations, this data point is discarded. So,

$$\boxed{\boldsymbol{D_{avg}^{Gaussian} \approx 20.8433nm}} \tag{9}$$

4.2.2 Lorentzian data

Equation (7) used for analysis. Also, we used $\boldsymbol{\beta_{instrument}^{Lorentz} \approx 0.001673819\ rad}$ for the analysis.

$(fwhm)^\circ$	β_{total}	β_{cryst}	θ	$D(nm)$
0.54337	0.009483596	0.007809777	0.315652656	19.50671166
0.33446	0.005837428	0.00416361	0.27630255	36.15270308
0.6754	0.011787954	0.010114135	0.414125965	15.64032064
~~0.13402~~	~~0.00233909~~	~~0.000665272~~	~~0.395976548~~	~~235.9368349~~
0.59824	0.010441258	0.008767439	0.59224083	19.90802377
0.53395	0.009319186	0.007645367	0.299618927	19.82494045
0.70504	0.012305269	0.010631451	0.547694704	15.95534127
0.55456	0.009678898	0.008005079	0.493043209	20.53653634

Here we can see clearly at $\boldsymbol{\theta \approx 0.396\ rad}$ we are getting $\boldsymbol{D \approx 235.93\ nm}$, which is exceptionally high. So, for further calculations, this data point is discarded. So,

$$\boxed{\boldsymbol{D_{avg}^{Lorentz} \approx 21.0749\ nm}} \tag{10}$$

Conclusion

From the above plot we can see Lorentz fit gives the optimal results. So, for this section we will consider $\boldsymbol{D_{avg}^{Lorentz} \approx 21.0749\ nm}$ as **Average Crystal Size**.

4.3 Williamson-Hall plot

To take the microstrain into account, we need to get the value of ϵ from the Williamson-Hall plot ignoring the data for the peak having erroneous results for the 'd' value. From the slope of the fitted line, we can infer that strain parameter.

Let's have,

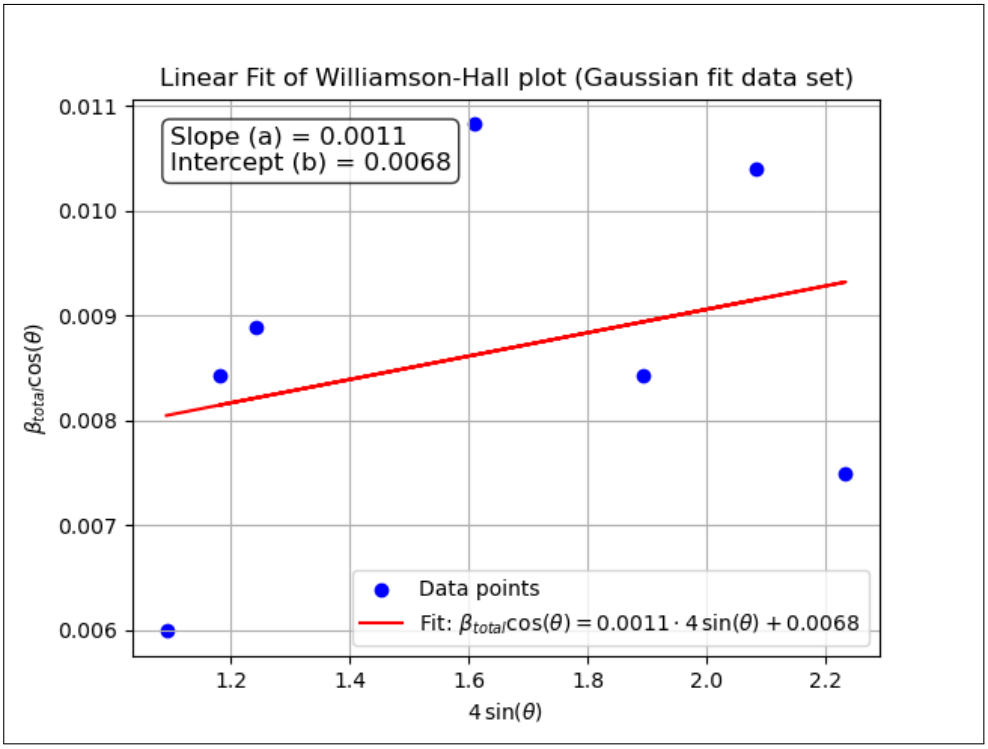
$$\beta_{total} = \beta_{crystal} + \beta_{strain} \tag{11}$$

From Equation (4) & (5) we obtain,

$$\beta_{total} \cos\theta = \frac{k\lambda}{D} + 4\epsilon \sin\theta \tag{12}$$

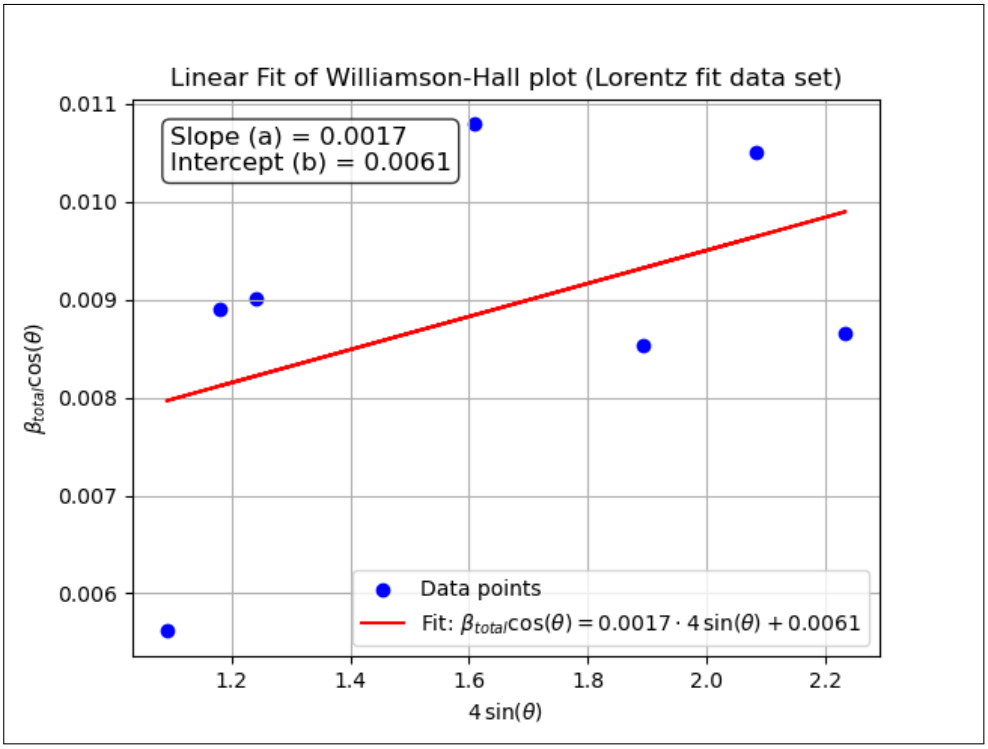
Using equation (12), we can plot $\boldsymbol{\beta_{total} \cos\theta}$ **vs.** $\boldsymbol{4 \sin\theta}$ & from slope we can obtain $\boldsymbol{\epsilon}$ **(strain parameter).**

4.3.1 Gaussian Data set



$$\boxed{\boldsymbol{Strain Parameter \equiv \epsilon_{gaussian} \approx 0.0011}} \tag{13}$$

4.3.2 Lorentzian Data set



$$\boxed{\boldsymbol{StrainParameter \equiv \epsilon_{lorentzian} \approx 0.0017}} \tag{14}$$

5 Error Analysis

This section gives a fully explicit, step-by-step error analysis for the crystallite-size estimates obtained from X-ray diffraction peak broadening. All numerical values (per-peak D values, instrumental-broadening estimates, and the noted outlier) are taken from the provided XRD Data.

5.1 Notation and basic equation

We use the Scherrer equation in the form

$$D = \frac{K\,\lambda}{\beta\,\cos\theta}, \tag{15}$$

where

- D is the coherent crystallite size (length units consistent with λ),
- K is the shape factor (dimensionless, typically 0.89–1.1 depending on morphology),
- λ is the X-ray wavelength (e.g. Cu K$\alpha \approx 1.5403\,$Å),
- β is the FWHM of *crystallite broadening* in *radians* (instrumental broadening removed), and
- θ is the Bragg angle (in radians).

Instrumental broadening must be removed by deconvolution. If the experimentally measured FWHM is β_{tot} and instrumental FWHM is β_{inst}, for Gaussian shapes one usually uses

$$\beta_{\text{cryst}} = \sqrt{\beta_{\text{tot}}^2 - \beta_{\text{inst}}^2},$$

whereas for Lorentzian profiles a subtraction in the half-widths or use of pseudo-Voigt formula is required. The report provides the instrumental terms used in the analysis:

$$\beta_{\text{inst}}^{(\text{Gaussian})} \approx 0.004574212 \text{ rad}, \qquad \beta_{\text{inst}}^{(\text{Lorentzian})} \approx 0.001673819 \text{ rad}.$$

5.2 Given per-peak crystallite sizes (after deconvolution)

After removing an anomalous outlier peak (at $\theta \approx 0.396$ rad, see report) the extracted per-peak crystallite sizes are:

Lorentzian-derived D (nm):

$$\{\, 19.5067,\ 36.1527,\ 15.6403,\ 19.9080,\ 19.8249,\ 15.9553,\ 20.5365 \,\}.$$

Gaussian-derived D (nm):

$$\{\, 35.6112,\ 18.6782,\ 14.5060,\ 22.4159,\ 20.0928,\ 15.0227,\ 19.5762 \,\}.$$

These per-peak numbers and the discarded outlier are listed in the report tables.

5.3 Statistical uncertainty from inter-peak dispersion (empirical approach)

Because the peak-fitting output did not include per-peak standard errors for β and θ in the provided tables, we first estimate the statistical uncertainty of the mean crystallite size empirically from the dispersion of the per-peak D values. This provides a conservative empirical uncertainty that includes peak-shape, background and overlap effects implicitly.

We compute sample statistics for the Lorentzian set (preferred model in the report) with $N = 7$ peaks.

Step 1: sum and mean

$$\sum_{i=1}^{N} D_i \;=\; 19.5067{+}36.1527{+}15.6403{+}19.9080{+}19.8249{+}15.9553{+}20.5365 \;=\; 147.5244$$

$$\overline{D}_{\text{L}} \;=\; \frac{1}{N}\sum_{i=1}^{N} D_i \;=\; \frac{147.5244}{7} \;=\; 21.0749142857 \text{ nm}.$$

Step 2: sample sum of squared deviations Compute squared deviations about the mean and sum:

$$\text{SS} \;=\; \sum_{i=1}^{N} (D_i - \overline{D})^2 \;=\; 288.7585155686.$$

(Intermediate arithmetic: e.g. $(19.5067 - 21.074914)^2 \approx 2.4570$, etc.; the full list of squared deviations sums to the SS value above.)

Step 3: sample variance and sample standard deviation

$$s^2 = \frac{\mathrm{SS}}{N-1} = \frac{288.7585155686}{6} = 48.1264192614,$$
$$s = \sqrt{s^2} = \sqrt{48.1264192614} = 6.9373207553 \text{ nm}.$$

Step 4: standard error of the mean (SEM)

$$\mathrm{SE}(\overline{D}_{\mathrm{L}}) = \frac{s}{\sqrt{N}} = \frac{6.9373207553}{\sqrt{7}} = 2.6220607834 \text{ nm}.$$

Thus, from inter-peak dispersion alone,

$$\overline{D}_{\mathrm{L}} = 21.0749 \text{ nm} \quad (\mathrm{SE} = 2.6221 \text{ nm}).$$

For completeness, the Gaussian set yields:

$$\overline{D}_{\mathrm{G}} = 20.8433 \text{ nm}, \qquad s_{\mathrm{G}} = 7.0882666839 \text{ nm}, \qquad \mathrm{SE}_{\mathrm{G}} = 2.6791129817 \text{ nm}.$$

5.4 Formal propagation of measurement uncertainties (analytic method)

The Scherrer equation, Eq. (15), depends on several measured/assumed quantities; to propagate uncertainties, we use partial-derivative (first-order) propagation.

If $D = f(K, \lambda, \beta, \theta)$, then the variance of D (assuming independent errors) is approximately

$$\sigma_D^2 = \left(\frac{\partial D}{\partial K}\sigma_K\right)^2 + \left(\frac{\partial D}{\partial \lambda}\sigma_\lambda\right)^2 + \left(\frac{\partial D}{\partial \beta}\sigma_\beta\right)^2 + \left(\frac{\partial D}{\partial \theta}\sigma_\theta\right)^2.$$

Compute the partial derivatives from Eq. (15):

$$\frac{\partial D}{\partial K} = \frac{\lambda}{\beta\cos\theta}, \qquad \frac{\partial D}{\partial \lambda} = \frac{K}{\beta\cos\theta},$$
$$\frac{\partial D}{\partial \beta} = -\frac{K\lambda}{\beta^2\cos\theta} = -\frac{D}{\beta}, \qquad \frac{\partial D}{\partial \theta} = \frac{K\lambda\sin\theta}{\beta\cos^2\theta} = D\,\tan\theta.$$

Substituting these into the variance expression and dividing both sides by D^2 gives the fractional variance:

$$\left(\frac{\sigma_D}{D}\right)^2 = \left(\frac{\sigma_K}{K}\right)^2 + \left(\frac{\sigma_\lambda}{\lambda}\right)^2 + \left(\frac{\sigma_\beta}{\beta}\right)^2 + (\sigma_\theta\,\tan\theta)^2,$$

which is the same result obtained by logarithmic differentiation.

Interpretation of terms

- σ_K/K is the fractional uncertainty in the Scherrer shape factor.
- σ_λ/λ is typically negligible for a monochromated Cu Kα source (unless the wavelength was poorly specified).
- σ_β/β is often the dominant term; σ_β should come from the peak-fit covariance (typical outputs of fitting software).
- σ_θ is the angular-position uncertainty (in radians); its contribution scales like $\tan\theta$ and becomes more important at higher angles.

5.5 Combining statistical and systematic uncertainties

A practical reported uncertainty must include both statistical (random) uncertainties and dominant systematic uncertainties (e.g. uncertainty in K, inadequately characterized instrumental profile). Combine independent contributions in quadrature:

$$\sigma_{\text{tot}} \;=\; \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{sys}}^2}.$$

Using the Lorentzian empirical statistics:

$$\overline{D}_{\text{L}} = 21.0749 \text{ nm}, \qquad \sigma_{\text{stat}} = \text{SE} = 2.6221 \text{ nm}.$$

Assume a conservative $\pm 5\%$ systematic uncertainty in K:

$$\sigma_{\text{sys},K} = 0.05 \times 21.0749 \;=\; 1.0537 \text{ nm}.$$

Thus the combined uncertainty is

$$\sigma_{\text{tot}} = \sqrt{2.6221^2 + 1.0537^2} \;=\; 2.8268 \text{ nm}.$$

If one instead assumes a $\pm 10\%$ uncertainty in K, $\sigma_{\text{sys},K} = 2.1075\,\text{nm}$ and

$$\sigma_{\text{tot}} = \sqrt{2.6221^2 + 2.1075^2} \approx 3.356 \text{ nm}.$$

5.6 Summary of uncertainties

- Empirical (dispersion-based) result (preferred given available data):

$$\boxed{\overline{D}_{\text{L}} = 21.0749 \pm 2.6221 \text{ nm (stat.)}}$$

5.7 Limitations and recommended actions to reduce uncertainty

1. **Obtain and propagate per-peak fit errors.** Export σ_{β_i} and σ_{θ_i} from your peak-fitting software (Origin or equivalent) and use the analytic propagation (see Subsection above) to compute σ_{D_i}. Then use the weighted mean.

2. **Report units and conversions consistently.** Ensure β and θ are in *radians* during propagation; convert λ and D consistently (e.g. Å $\leftrightarrow$ nm).

6 Conclusion

- From the Lorentzian peak-fit dataset (after discarding the outlier at $\theta \approx 0.396\,\text{rad}$) we obtain an empirical average coherent crystallite size

$$\boxed{\overline{D}_{\text{L}} = 21.0749 \pm 2.6221 \text{ nm (stat.)}}$$

  where the statistical uncertainty is the standard error of the mean computed from the per-peak dispersion. The numerical values are taken directly from the report tables.

- The Gaussian-derived average is consistent:

$$\overline{D}_{\mathrm{G}} = 20.8433 \pm 2.6791 \text{ nm (stat.)},$$

  showing agreement (within uncertainties) between the two fit-model approaches; the report judged Lorentzian fits to better describe the observed peak shapes, hence $\overline{D}_{\mathrm{L}}$ is recommended as the principal reported value.

- The single anomalous value $D \approx 235.9\,\mathrm{nm}$ (at $\theta \approx 0.396\,\mathrm{rad}$) is inconsistent with the remainder of the dataset and was correctly discarded; plausible causes include poor local baseline subtraction, unresolved overlap or a unit/angle-conversion artefact (the report notes this peak as an outlier).